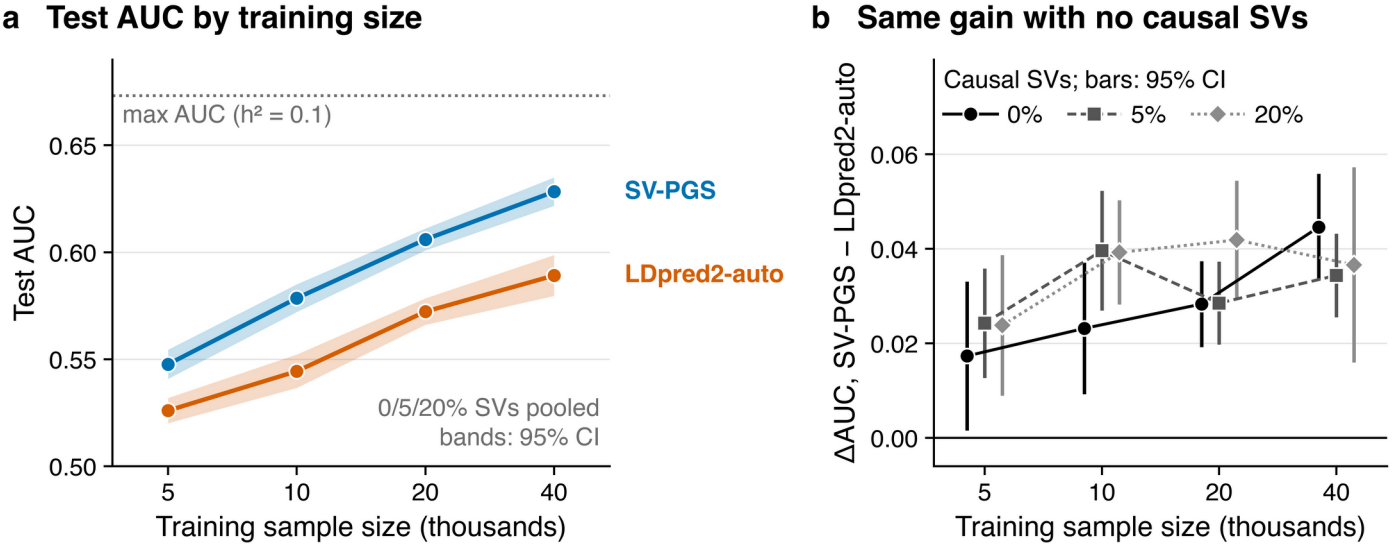


Figure 1 | SV-PGS is more accurate than LDpred2-auto at every training size, by the same margin whether 0%, 5% or 20% of the causal variants are SVs, so the gain does not come from causal SVs. A simulated binary trait on chromosome 22 (liability h^2 = 0.1, prevalence 10%, 200 causal variants) scored in the same 10,000 held-out people; SV-PGS fits every SNV, indel and imputed SV, LDpred2-auto HapMap3-density SNVs of the same training people. a, Test AUC, mean and Student-t 95% interval over the three settings' replicates pooled (42, 37, 38, 16 at 5k-40k); dotted, the largest AUC any score can reach, that of the true genetic value at h^2 = 0.1 (Wray et al. 2010). b, Paired difference in AUC on the same held-out people, per setting, mean and Student-t 95% interval.

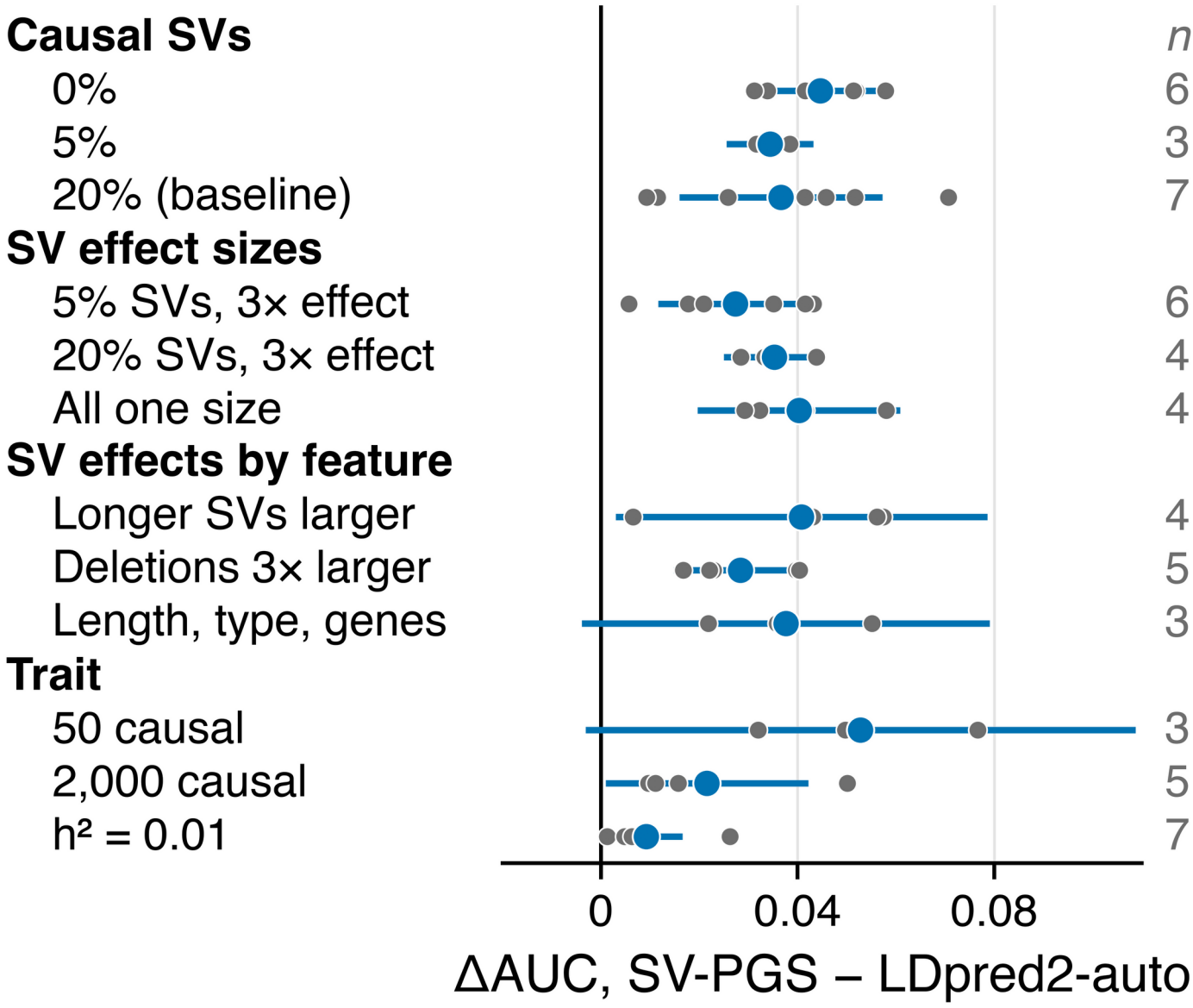


Figure 2 | SV-PGS is more accurate than LDpred2-auto across simulated genetic architectures. Paired difference in test AUC on the same 10,000 held-out people between SV-PGS (every SNV, indel and imputed SV) and LDpred2-auto (HapMap3-density SNVs), both trained on the same 40,000 people: replicates (grey), mean and Student-t 95% interval (blue; an arrowhead marks one that runs past 0.11), n replicates at right. SV-PGS had the higher AUC in 57 of 57 replicates. The baseline architecture has liability $h^2 = 0.1$, prevalence 10%, 200 causal variants and 20% causal SVs with SV effects drawn like small-variant effects, and every other changes one thing of it: '3×' triples the SV effect SD, 'a few very large' draws SV effects from a Student t with 3 degrees of freedom, 'all one size' gives every causal SV the same effect size with a random sign, 'well-imputed' draws causal SVs only from those imputed with $r^2 \geq 0.8$, and 'common' only from SVs of frequency ≥ 5%. Not yet replicated three times at 40,000: a few very large SV effects; only well-imputed SVs; only well-imputed SVs, 3× effect; only common SVs; prevalence 1%.

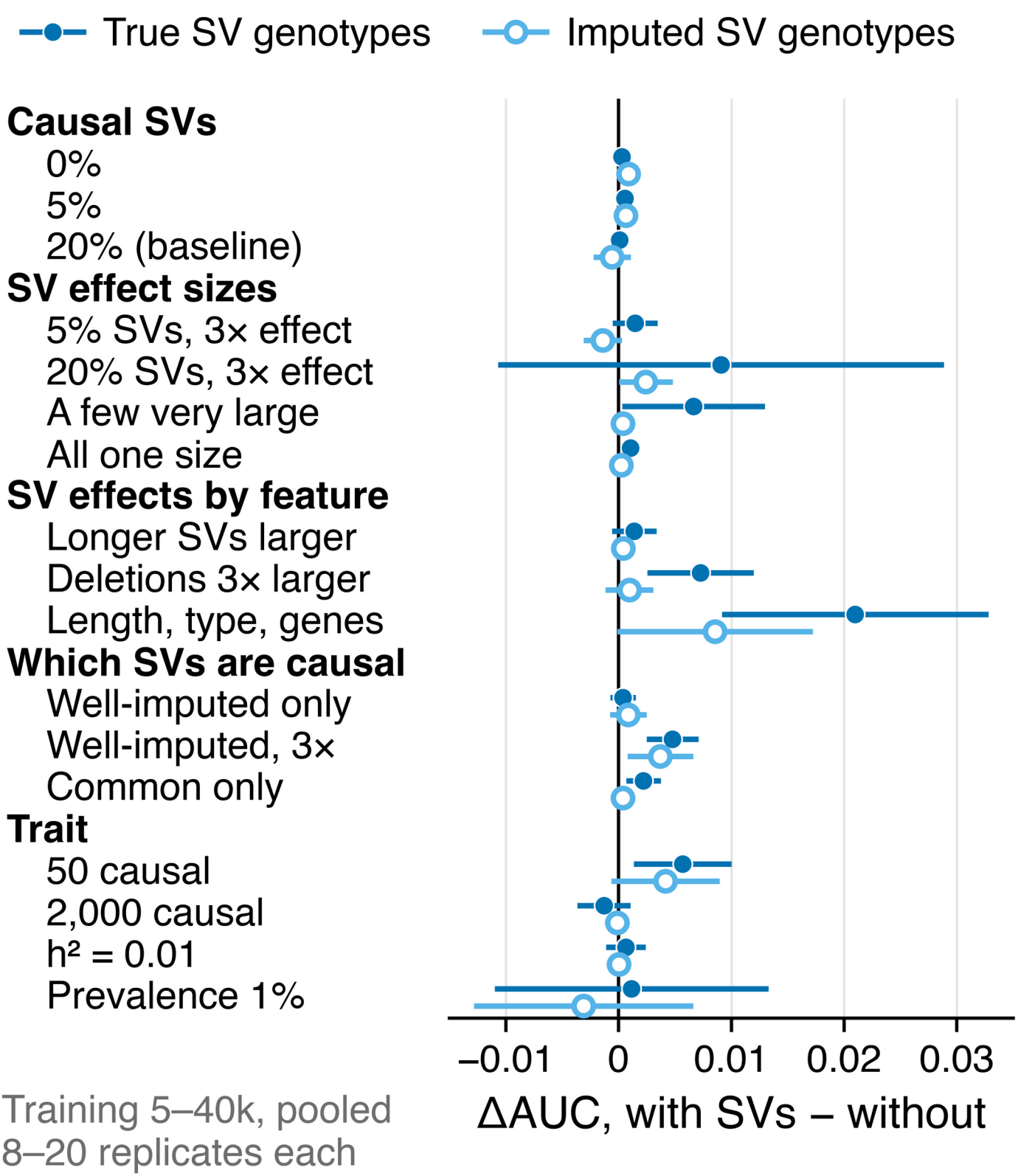


Figure 3 | SVs add accuracy when their effects are large and they are measured well; imputation removes most of the gain. Paired difference in test AUC on the same held-out people between SV-PGS fitted with and without SVs on one 3 Mb chromosome 22 window (60,000 records), with every SV measured by its true genotype (filled) or imputed by Beagle from SNV haplotypes (open): mean and Student-t 95% interval over 8-20 replicates, every training size from 5,000 to 40,000 pooled (Figure 4 shows them apart). The true-genotype interval lies above zero for 7 of 17 architectures. Same simulated people, truth and fits; only the SV genotypes differ. The widest true-genotype interval, 20% SVs, 3× effect, comes from one replicate at –0.085. The baseline architecture has liability $h^2 = 0.1$, prevalence 10%, 200 causal variants and 20% causal SVs with SV effects drawn like small-variant effects, and every other changes one thing of it: '3×' triples the SV effect SD, 'a few very large' draws SV effects from a Student t with 3 degrees of freedom, 'all one size' gives every causal SV the same effect size with a random sign, 'well-imputed' draws causal SVs only from those imputed with $r^2 \geq 0.8$, and 'common' only from SVs of frequency ≥ 5%.

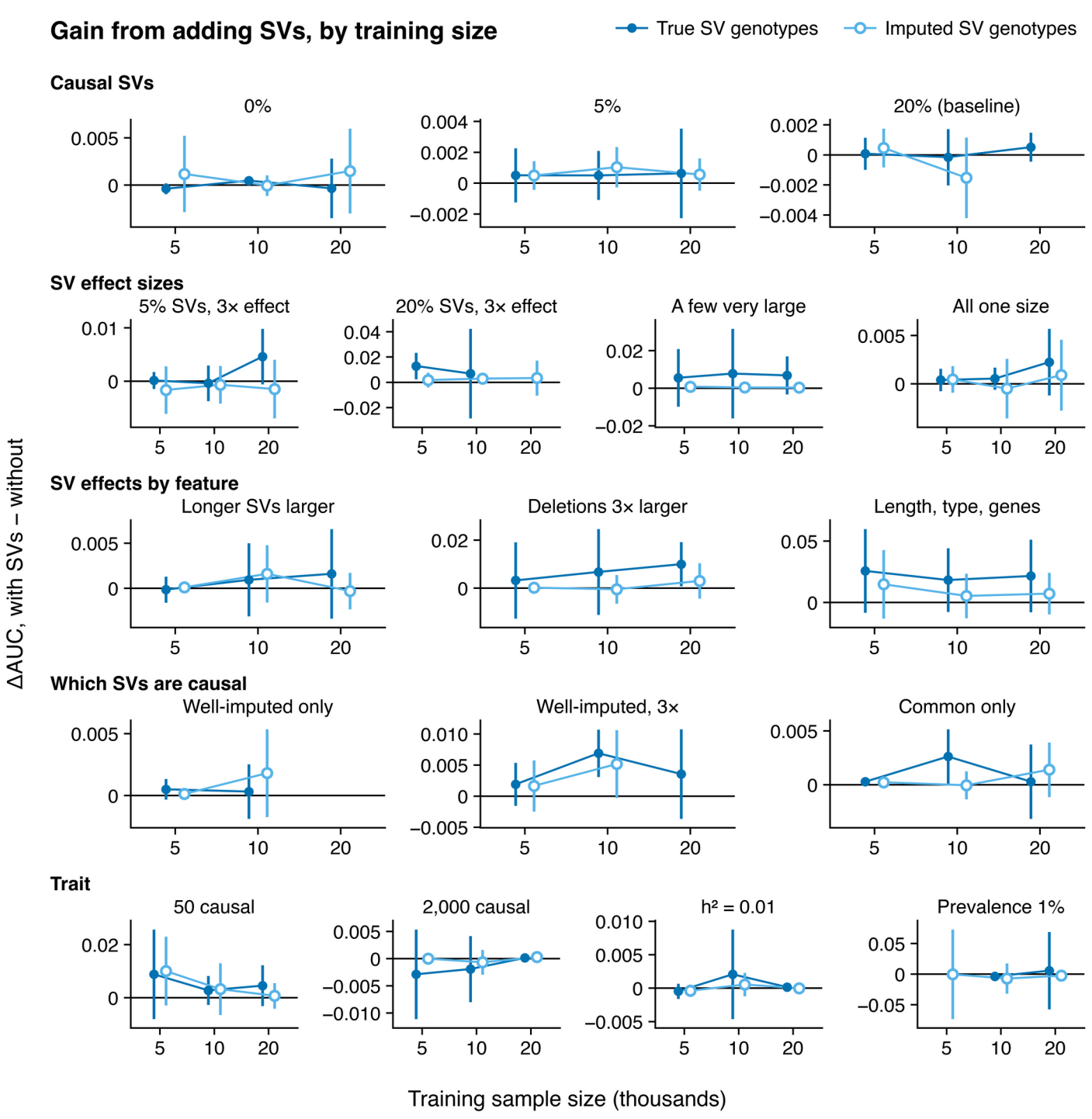


Figure 4 | The gain from adding SVs at each training size from 5,000 to 20,000, for each simulated architecture. Each panel shows the paired difference in test AUC between SV-PGS fitted with and without SVs on one 3 Mb chromosome 22 window, with SVs measured by their true genotypes (filled) or imputed by Beagle (open): the mean over replicates and its Student-t 95% interval. Each panel has its own y-axis scale. A mean is drawn when its training size has at least three replicates. Each replicate uses the same simulated truth and the same test people at every training size. We fitted 0-3 replicates per architecture and genotype source at 40,000 training people, too few to show; Figure 3 pools them with the other sizes. The baseline architecture has liability $h^2 = 0.1$, prevalence 10%, 200 causal variants and 20% causal SVs with SV effects drawn like small-variant effects, and every other changes one thing of it: '3×' triples the SV effect SD, 'a few very large' draws SV effects from a Student t with 3 degrees of freedom, 'all one size' gives every causal SV the same effect size with a random sign, 'well-imputed' draws causal SVs only from those imputed with $r^2 \geq 0.8$, and 'common' only from SVs of frequency ≥ 5%.

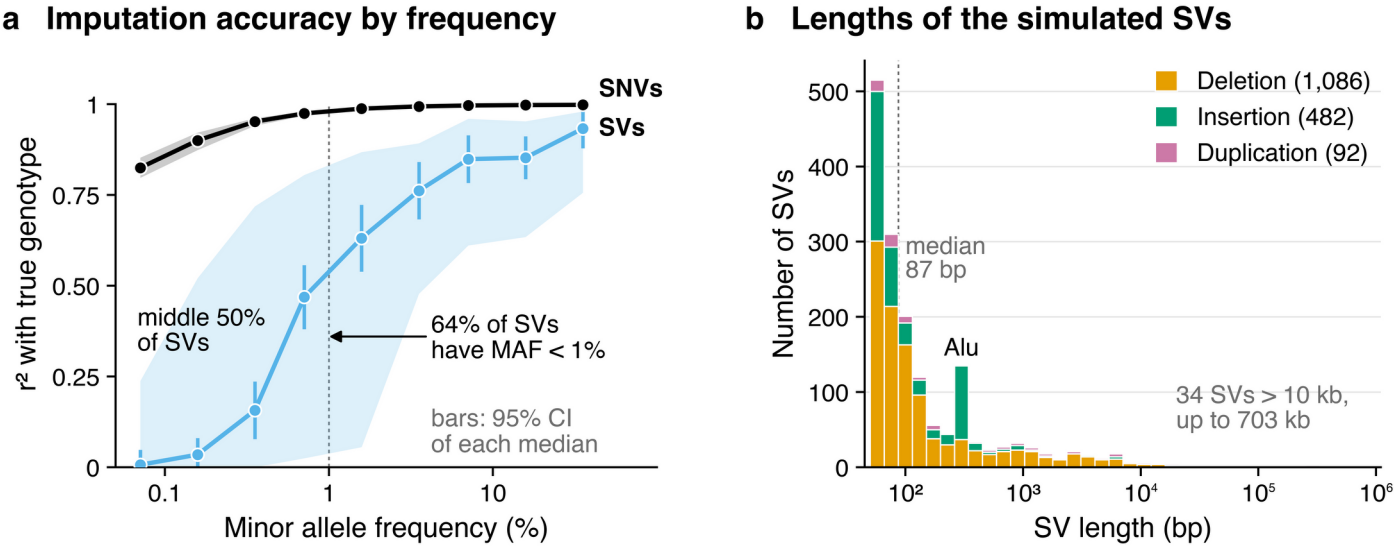


Figure 5 | The simulated SVs are mostly rare and short, and rare SVs impute poorly. a, The r^2 of each variant's measured dosage with its true genotype over all 50,000 simulated people (not Beagle's DR2), for 1,662 SVs imputed by Beagle and 6,000 sampled SNVs called from simulated reads on chromosome 22, by minor allele frequency: each bin's median (points) with an approximate 95% interval (1.82 IQR/$\sqrt{n}$; bars), over the bin's middle 50% of variants (bands; SNVs grey, SVs blue). b, Lengths of the 1,662 measured SVs, stacked by class (records without a type are classed by the sign of their length change); the insertions' peak near 300 bp is where Alu elements lie. bench-sim v7 cohort from public 1000 Genomes haplotypes.

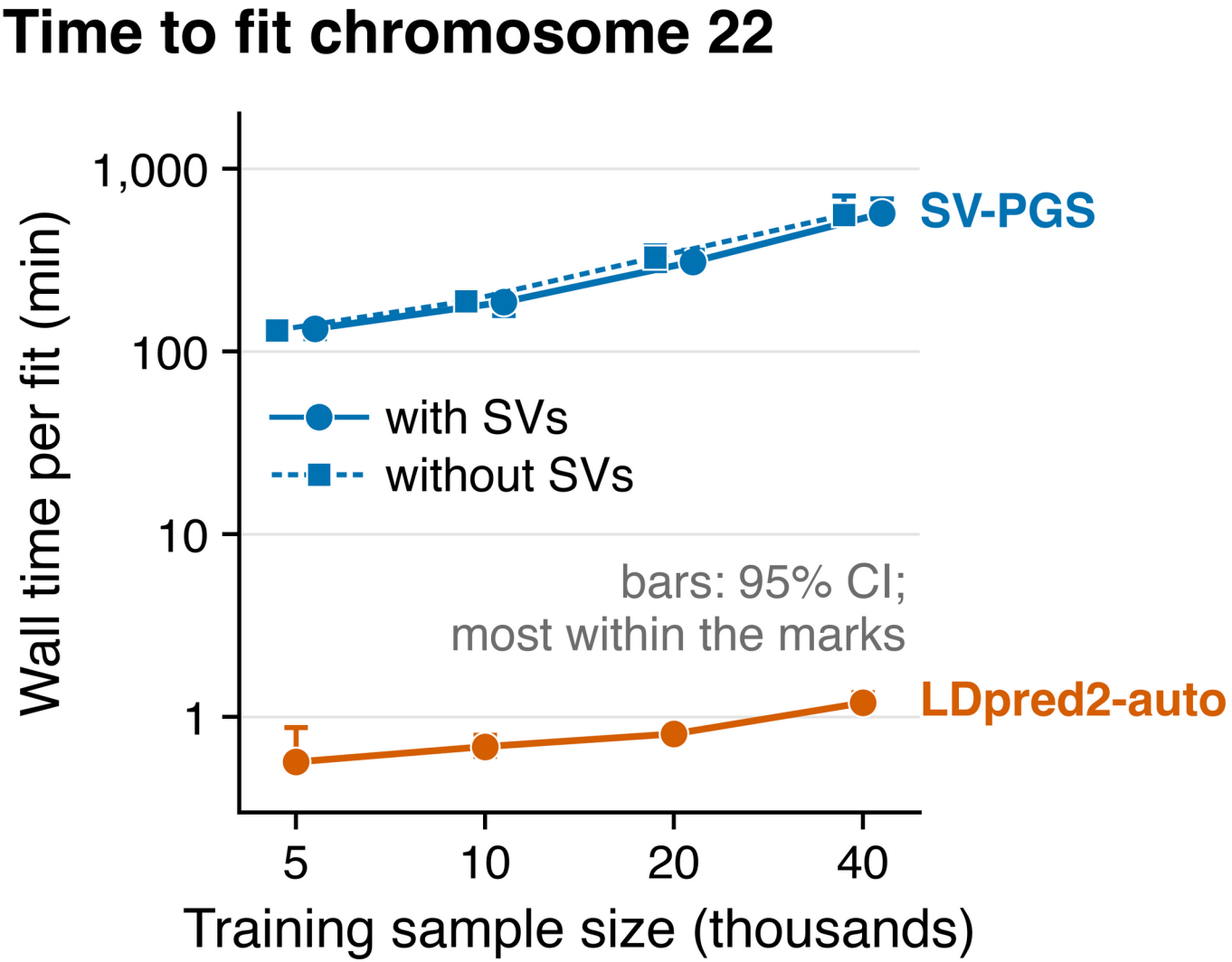


Figure 6 | SV-PGS fits take hours and LDpred2-auto's minutes; SVs add no time. Wall time of one fit on chromosome 22 per method and training size, pooled over every architecture: median and its distribution-free 95% interval (59-87 fits each; capped bars, most within the marks). MSI Slurm jobs, one fit per job; SV-PGS fits every variant by empirical Bayes (circles and solid line, with SVs; squares and dashed line, without), LDpred2-auto HapMap3-density SNVs. The results do not record each job's hardware, so the gap compares the methods as they were run, not on equal resources.